

SIDONG ZHANG

✉ sidongzhang@umass.edu

in [linkedin.com/in/sidong-zhang-b25502143](https://www.linkedin.com/in/sidong-zhang-b25502143)

🏠 <https://stonezhng.github.io>

Education

UMass Amherst's College of Information and Computer Sciences

Amherst, MA, USA

Doctor of Philosophy

Sep. 2020 - present

- Teaching assistant of graduate level CS 589 machine learning and CS 651 optimization class
- Research assistant in Information Fusion lab, currently funded by an NIH RO3

UMass Amherst's College of Information and Computer Sciences

Amherst, MA, USA

Master of Science

Sep. 2018 - Jan. 2021

- GPA: 3.82/4.0

Nanjing University, Software Institute

Nanjing, China

Bachelor of Engineering

Sep. 2014 - June 2018

- GPA: 4.35/5.0

Project

Audio-Visual Speech Separation via Bottleneck Iterative Network

Accepted to ICML 2025 Workshop on Machine Learning for Audio

Feb. 2024 - Sep. 2024

- We work on audio-visual speech separation on noisy audio mixtures of NTCD-TIMIT and LRS3 data set
- We propose a novel multimodal fusion framework on audio and video modality with a bottleneck iterative structure
- Our proposed model outperforms state-of-the-art on SI-SDRi and only requires 50% of SOTA training time

Longitudinal Multimodal Modeling for Alzheimer's Early Detection in the Wild

Submitting to Medical Image Analysis

Sep. 2020 - July 2025

- We introduce information theory-based unsupervised representation learning on brain MRIs as a complement to risk factors, with an estimated mutual information value to evaluate the strength of the dependency to risk factors
- We both train CNN from scratch and finetune foundation models as two representation learning models
- We optimize the procedure of training Alzheimer's forecasting model to include two stages with partial model parameter freezing for stabler validation performance during training process
- We constructed a representative subset from the ADNI dataset aligned with the TADPOLE challenge, consisting of 1,183 patients for training, 169 for validation, and 336 for testing
- We evaluate the forecasting performance using both micro F1-score on CN-MCI-AD forecasting and the precision of capturing MCI to AD transition timing across 100 repeated experiments to ensure statistical significance

Encoding Domain Insights into Multi-modal Fusion: Improved Performance at the Cost of Robustness

Accepted to ICML 2025 Workshop on Methods and Opportunities at Small Scale, Jan. 2025 - May 2025

- We work on comparing fusion methods with and without domain knowledge embedded in the model structure
- We conduct experiments on MMSD dataset with both the original task of detecting sarcasm and a synthetic task to control the level of domain knowledge, with additional Gaussian noise added to the inputs to examine robustness
- We find that aligning fusion design with priors boosts clean-data accuracy with limited data but significantly diminishes robustness to noisy inputs.

Clustered Vertical Attention for Irregular Time Series Modelling

Submitted to ICML 2019 TimeSeriesWorkshop

Feb. 2019 - May 2019

- We worked on PhysioNet Challenge 2012 data set
- We improved the prediction accuracy of the in-hospital survival of ICU patients via existing imputation methods
- We ran Minimum Spanning Tree algorithm to determine the most correlated imputed clusters
- We trained separate Attention models for clusters and predicted on a Long Short-term Memory Attention model
- The model got accuracy improvements of 1.5%, 1.3% and 1.8% on 3 different imputation methods

Technical Skills

Languages: English (Proficient), Mandarin (Native)

Programming language: Java, Python, C, Lisp, Markdown, Latex

Research interests: multimodal learning, healthcare, computer vision